

COMP527 - CA Assignment 2 - Preprocessing

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1 Overview of the Pipeline

`clustering.py` calls the functions from `preprocessing.py` to call and clean the data and then uses TF-IDF vectorisation on the data to get a mathematical abstraction of the sentences. Latent Semantic Analysis was used to simplify the geometry of the resulting vector set. L2 normalisation was implemented as well.

This final set was then used to get the silhouette analysis.

2 Term Frequency - Inverse Document Frequency (TF-IDF)

TF-IDF is a statistical method that determines the relevance of a word in relation to the document it appears in. In this case, the term frequency for a single term t of a sentence s is described as:

$$TF = \frac{\text{Frequency of } t}{\text{Count of unique terms in } s}$$

The inverse document frequency is described mathematically as:

$$TF = \frac{\text{Total number of sentences}}{\text{Count of sentences with term } s}$$